6.0 Framework

In this chapter we will consider a framework consists of light rods joined at their ends to other rods with smooth joints.

6.1 Rigid Frame

If the shape of a frame is unaltered by external forces, then the frame in called a rigid frame.

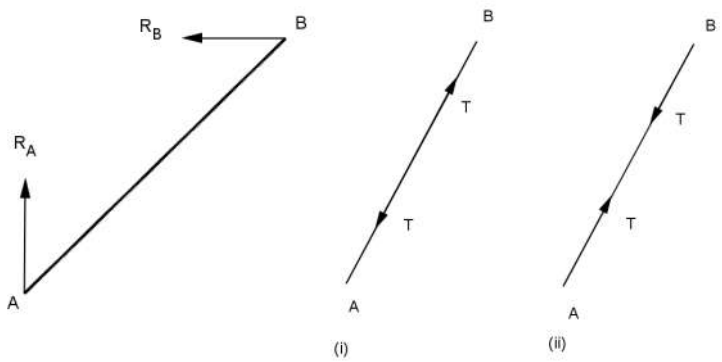
In a frame made by light rods, the reactions at the joints will act along the rods. These reactions along the rods are known as stresses.

If we consider a light rod AB in a frame R_A and R_B are the reactions at the joints by pins. The rod is in equilibrium under the action of these two forces R_A and R_B. Hence for the equilibrium of the rods R_A and R_B must be equal and act opposite along the rod.

$R_A = R_B = T$

(i) T is tension

(ii) T is thrust



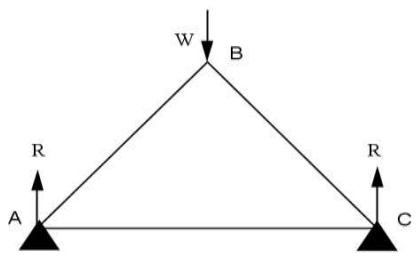
Assumptions when solving framework problems

- All the rods in the framework are light rods.
- All the rods are freely (smoothly) jointed at their ends and no couple in formed at a joint.
- The reactions at the joints (except external forces) will act along the rods. These may be thrusts or tensions.
- All the rods in the frame are in the same vertical plane and all the forces (including the external forces) are coplanar forces.
- External forces are applied only on joints.

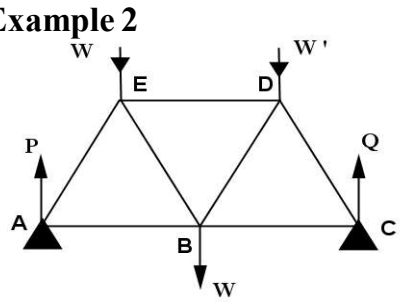
6.2 Representing external forces in a light framework in equilibrium

Example 1

Δ ABC is frame lying on A and C, carries a load *W* at B. By symmetry the reactions at A and C are equal.



Example 2



ABCDE is a frame made of seven equal light rods and rests on two pegs at A and C. It carries *W* at E, B and W′ at D. The external forces P, Q will be vertical.

Bow's Notation

- This notation is introduced by a mathematician called Bow.
- All the external forces will be represented outside of the frame.
- The region between forces (open or closed) is denoted by a small letter of the English alphabet or a number.
- Each force denoted by two letters of the alphabet belongs to the two region formed by the force.

Solving problems using Bow's notation

(i) Having represented all external forces and regions, forces of polygons have to be drawn for each joint of the frame (These polygon of forces will be a closed figure, the vertices of the polygon being denoted by the names of the letters of the regions.)

(ii) The values of the stresses in rods can be calculated by using trigonometric ratios and algebraic equations in the triangles and polygons obtained in the stress diagram.

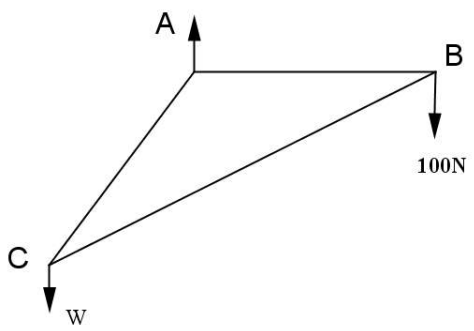
(iii) By reading the names of the sides in the stress diagram, mark the directions of the stress by using arrow marks.

(iv) While drawing force polygons, the disense has to be same for all the joints. (either clockwise or anticlockwise)

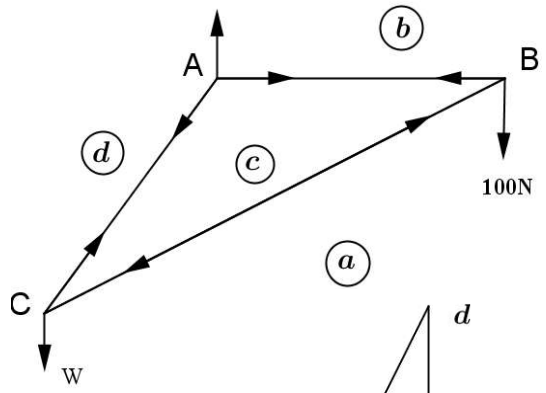
(v) To draw a polygon of forces at a joint there may be maximum of two unknown forces.

6.3 Worked examples

Example 1

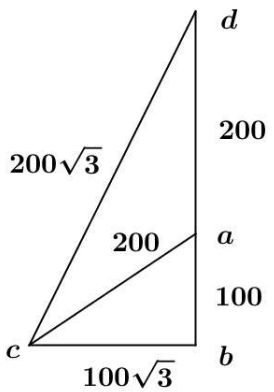


In the given figure, ABC is a triangular framework consisting of three smoothly jointed light rods AB, BC, CA, where AB = AC and $B\hat{A}C=120°$. The framework is in a vertical plane with AB horizontal. It is supported at A by a smooth peg and carries loads 100 N at B and W N at C. Draw a stress diagram using Bow's notation and from it, calculate the stresses in the rods, distinguishing between tensions and thrusts and also find the value of W.



Start from joint B

Joint	Order	Name of Polygon
B	$a \to b \to c \to a$	Δabc
C	$a \to c \to d \to a$	Δbcd



AB(bc) = Tension = $100\sqrt{3}$ N

BC(ca) = Thrust = $200\sqrt{3}$ N

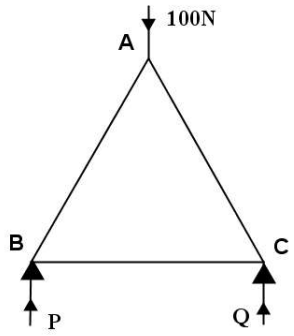
CA(cd) = Tension = $200\sqrt{3}$ N

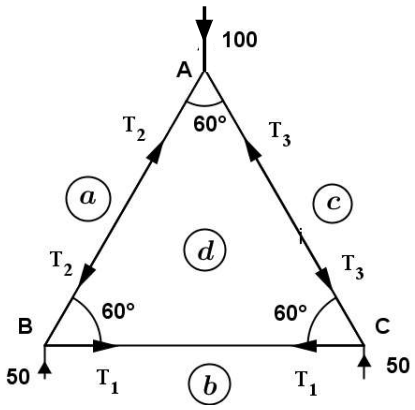
W(ad) = 200 N

In this problem all the joints are taken in the anticlockwise sense.

Example 2

ABC is a frame obtained by joining three uniform equal light rods AB, BC and AC. B and C rest on 2 pegs at the same horizontal level. A carries a load of 100N. Find the reaction at B and C. Draw a stress diagram by using Bow's notation. Hence find the stress in each rod distinguishing between tension and thrust.





For equilibrium

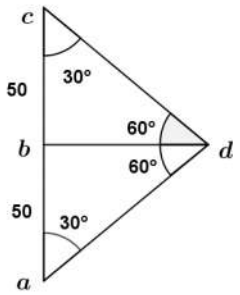
Resolve the forces vertically

$\uparrow P + Q = 100$

$P = Q = 50$ (symmetry)

Polygon of forces has to be drawn for joints A, B and C by naming the regions between the vertices as a,b,c and d.

Stress diagram



This diagram is drawn by taking the region around each joint in anticlockwise disense starting from C.

Joint C $\rightarrow$ Joint A $\rightarrow$ Joint B

Joint	order	Name of Polygon
C	$b \rightarrow c \rightarrow d \rightarrow a$	Δbcd
A	$d \rightarrow c \rightarrow a \rightarrow d$	Δacd

Tensions and thrusts are denoted by naming the regions.

$T_1 = bd = 50 \tan 30° = \frac{50}{\sqrt{3}}$ N

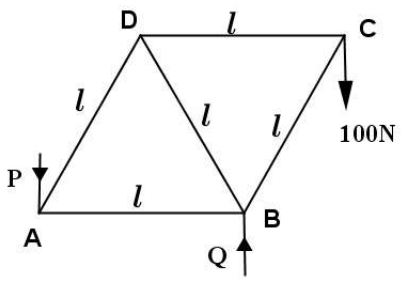
$T_3 = cd = 50 \sec 30° = \frac{100}{\sqrt{3}}$ N

$T_2 = ad = 50 \sec 30° = \frac{100}{\sqrt{3}}$ N

Rod	Stress	Thrust	Tension
AB	$\frac{100}{\sqrt{3}}$ N	✓	-
BC	$\frac{50}{\sqrt{3}}$ N	-	✓
AC	$\frac{100}{\sqrt{3}}$ N	✓	-

Example 3

The given figure represents the framework of five equal light rods. This frame is supported by a peg at B and a vertical force P is applied at A. C carries a load of 100 N. Find the stresses in each rod by drawing a stress diagram.



For equilibrium

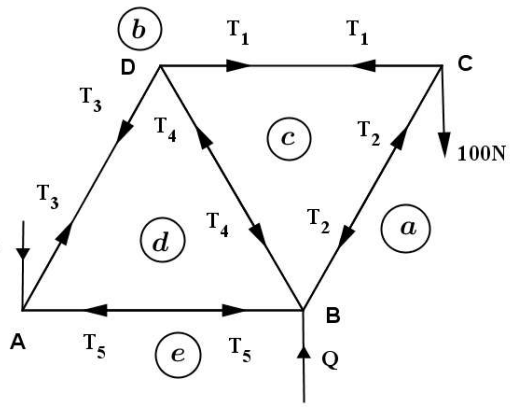
Resolve the forces vertically

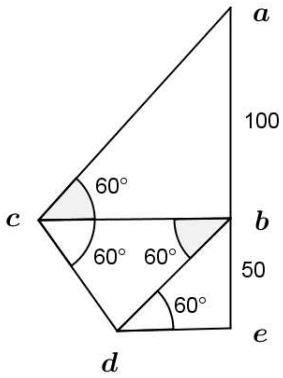
$\uparrow\ 100 + P = Q$ ..... ①

By taking moments about A

$Am \quad Q.2\ell = 100\,(2\ell + 2\ell\cos 60°)$ .....② P

① and ② $\Rightarrow$ P = 50 N , Q = 150 N





In the above diagram regions are named starting from C and the stress diagram is drawn as follows.

Joint	order	Name of Polygon
C	$a \to b \to c \to a$	Δabc
D	$c \to d \to c \to b$	Δbcd
A	$d \to b \to e \to d$	Δdbe
B	$c \to d \to e \to a \to c$	$\square acde$

The force polygon is drawn starting from joint C joining the region in the anticlockwise disense.

$T_1 = bc = 100 \tan 30° = \frac{100\sqrt{3}}{3}$ N

$T_2 = ac = 100 \sec 30° = \frac{200\sqrt{3}}{3}$ N

$T_3 = bd = 50 \operatorname{cosec} 60° = \frac{100\sqrt{3}}{3}$ N

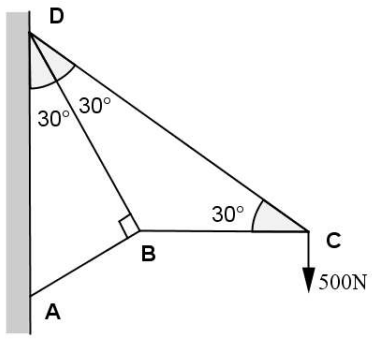
$T_4 = cd = bd = \frac{100\sqrt{3}}{3}$ N

$T_5 = dc = 50 \tan 30° = \frac{50\sqrt{3}}{3}$ N

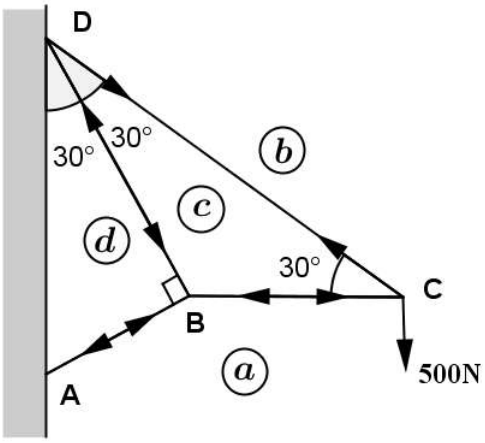
Rod	Stress	
DC	$\frac{100\sqrt{3}}{3}$ N	Tension
BC	$\frac{200\sqrt{3}}{3}$ N	Thrust
AD	$\frac{100\sqrt{3}}{3}$ N	Tension
BD	$\frac{100\sqrt{3}}{3}$ N	Thrust
AB	$\frac{50\sqrt{3}}{3}$ N	Thrust

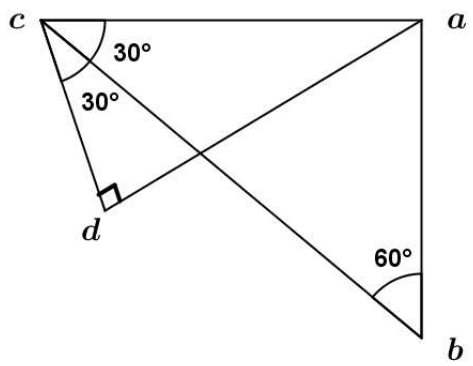
Example 4

A framework formed by four light rods AB, BC, CD and BD is shown in the given diagram. A, D are freely jointed to a vertical wall. Joint C carries a load of 500 N and BC remains horizontal. Draw a stress diagram using Bow's notation and find the stresses in each rod distinguishing between tensions and thrusts.



At joint C one force is known and two forces unknown. Draw the stress diagram starting from joint C.





Joint	order	Name of Polygon
C	$a \to b \to c \to a$	Δabc
B	$a \to c \to d \to a$	Δacd

bc = 500 sec 60° = 1 000 N

ac = 500 tan 60° = $500\sqrt{3}$ N

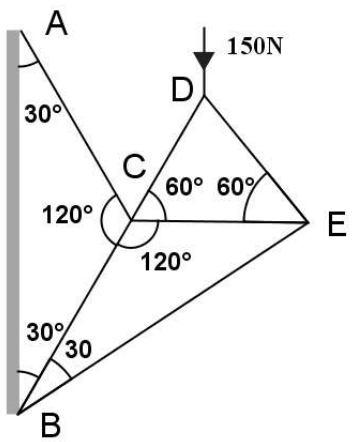
cd = ($500\sqrt{3}$ N) sin 30° = $250\sqrt{3}$ N

ad = $500\sqrt{3}$ N cos 30° = 750 N

Rod	Stress	Thrust	Tension
DC	1 000 N	-	✓
BC	$500\sqrt{3}$ N	✓	-
BD	$250\sqrt{3}$ N	✓	-
AB	750 N	✓	-

Example 5

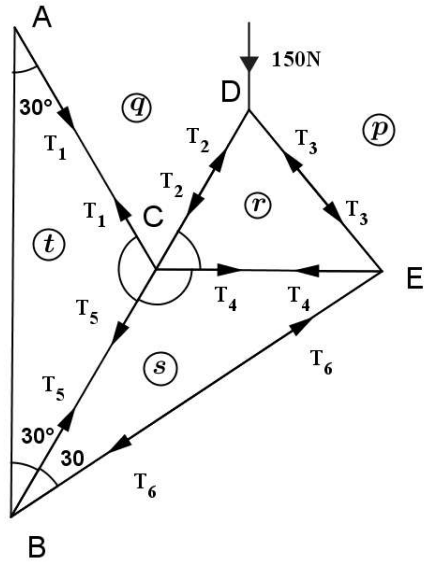
The given figure show a framework of six light rods smoothly jointed at C, D and E. A and B are smoothly jointed to a vertical wall and D carries a load of 150N. Draw a stress diagram using Bow's notation and find the stresses in each rod distinguishing between thrusts and tensions.

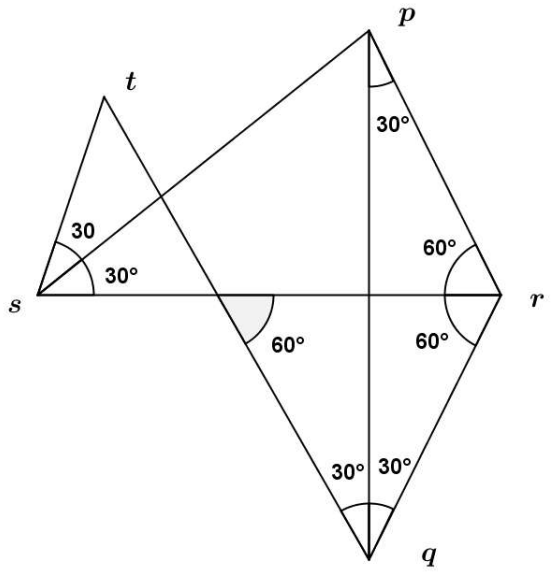


D is the joint with one known and two unknown forces

So start to draw triangle of forces from joint D

Joint	order	Name of Polygon
D	$p \to q \to r \to p$	Δpqr
E	$p \to r \to s \to p$	Δprs
C	$r \to q \to t \to s \to r$	$\square rqts$





AC = qt = $75\sqrt{3} + 25\sqrt{3} = 100\sqrt{3}$ N

CD = qr = 75 sec 30° = $50\sqrt{3}$ N

DE = pr = qr = $50\sqrt{3}$ N

CE = sr = $100\sqrt{3}$ N

BC = st = $50\sqrt{3}$ N

BE = ps = $150\sqrt{3}$ N

Rod	Stress	Thrust	Tension
AC	$100\sqrt{3}$ N	-	✓
CD	$50\sqrt{3}$ N	✓	-
DE	$50\sqrt{3}$ N	✓	-
CE	$100\sqrt{3}$ N	-	✓
BC	$50\sqrt{3}$ N	-	✓
CE	$150\sqrt{3}$ N	-	✓

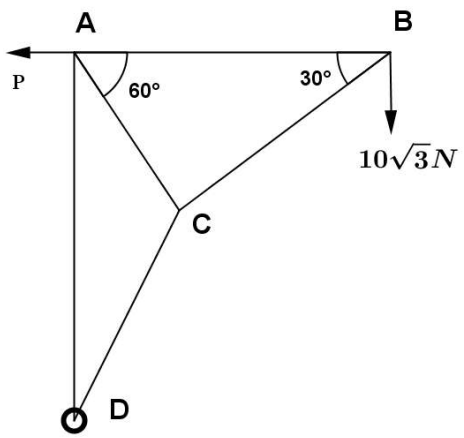
Example 6

Five rods AB, BC, CD, DA and AC are smoothly jointed at their ends to form a framework as shown in the figure. $A\hat{B}C = A\hat{D}C = D\hat{A}C = 30°$ and $B\hat{A}C = 60°$. The framework is smoothly hinged at D and carries a weight $10\sqrt{3}$ N at B. The framework is held in a vertical plane with AB horizontal by a horizontal force P at A.

(i) Find the value of P

(ii) Find the magnitude and direction of the reaction at D.

(iii) Using Bow's notation, draw a stress diagram for the framework and find the stresses in all the rods, distinguishing between tensions and thrusts.



(i) For equilibrium

Take moment about D

Dm P. AD - $10\sqrt{3}$ AB = 0

$$\text{but } AD = 2\,AC\cos 30°$$
$$= 2\,AB\cos 60°\cos 30°$$
$$AD = \frac{\sqrt{3}}{2}AB$$
$$\therefore\ P.\frac{\sqrt{3}}{2}AB = 10\sqrt{3}\,AB$$
$$\therefore\ P = 20\text{ N}$$

Let R be the reaction at D and θ be the angle that R makes with the horizontal

Resolving vertically

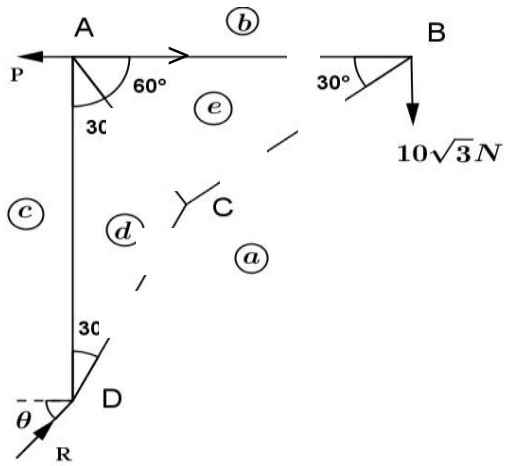
$\uparrow R\sin\theta = 10\sqrt{3}$

Resolving horizontally

$\rightarrow R\cos\theta = P = 20$ N

$$R = \sqrt{\left(10\sqrt{3}\right)^2 + 20^2} = 10\sqrt{7}$$

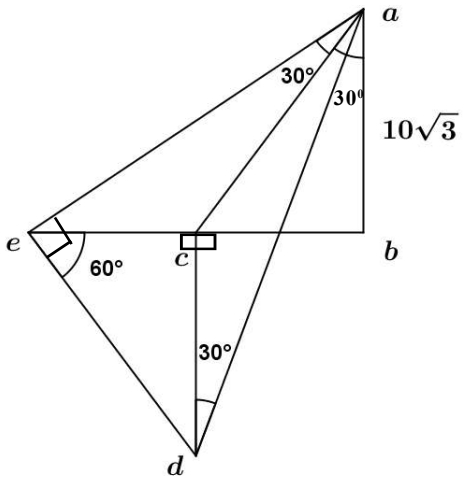
$$\tan\theta = \frac{10\sqrt{3}}{20} = \frac{\sqrt{3}}{2}\ ;\qquad \theta = \tan^{-1}\frac{\sqrt{3}}{2}$$



Since the system is in equilibrium under three forces the reaction R should also pass through B.

Start from joint B in the anticlockwise direction.

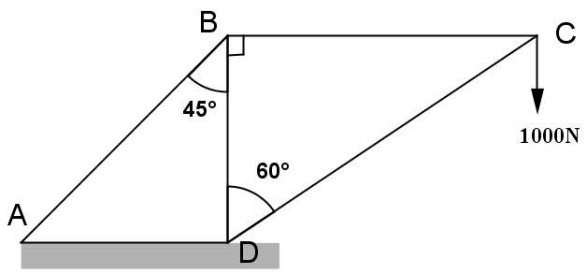
Joint	order	Name of Polygon
B	$a \rightarrow b \rightarrow e \rightarrow a$	Δabd
C	$a \rightarrow e \rightarrow d \rightarrow a$	Δaed



Rod	Stress	Magnitude
AB	Tension	30 N
BC	Thrust	$20\sqrt{3}$ N
AC	Thrust	20 N
DC	Thrust	40 N
AD	Tension	$10\sqrt{3}$ N

$$be = 10\sqrt{3}\tan 60 = 30$$
$$ae = 10\sqrt{3}\sec 60 = 20\sqrt{3}$$
$$ad = 20\sqrt{3}\sec 30 = 40$$
$$ed = ae\tan 30 = 20 \qquad cd = ed\sin 60 = 10\sqrt{3}$$

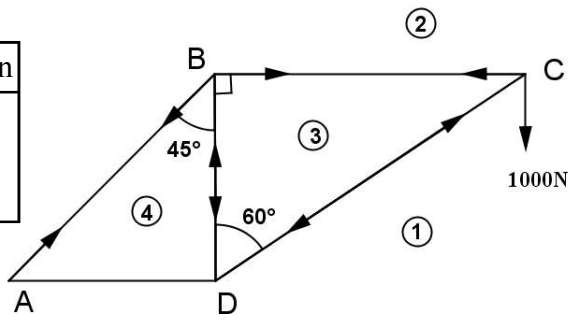
Example 7



The given figure shows a crane composed of four freely jointed rods AB, BC, CD and BD. The rod BC is horizontal while the rod BD is vertical. The crane is fixed to the horizontal ground at A and D and there is a load of 1 000 N hanging at C. Use Bow's notation to find the forces in the rods, distinguishing between tensions and thrusts.

Start with joint C in anticlockwise

Joint	order	Name of Polygon
C	$1 \to 2 \to 3 \to 1$	$\Delta 123$
B	$3 \to 2 \to 4 \to 3$	$\Delta 324$



$AB = km = 1000\sqrt{6}$ N

$BC = kl = 1\,000 \cot 30^0 = 1\,000\sqrt{3}$ N

$CD = lj = 1\,000 \text{ cosec } 30° = 2\,000$ N

$BD = ml = kl = P.\ell \cos 30° - 10\sqrt{3}\,.\,2\ell = 0$

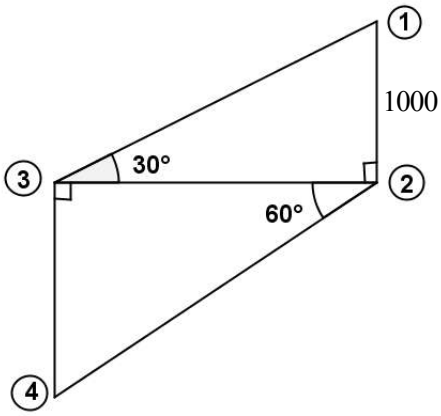
$\Rightarrow P = 40$ N

$\rightarrow\ P = R\cos\theta = 40$ N

$\uparrow R\sin\theta = 10\sqrt{3}$ N

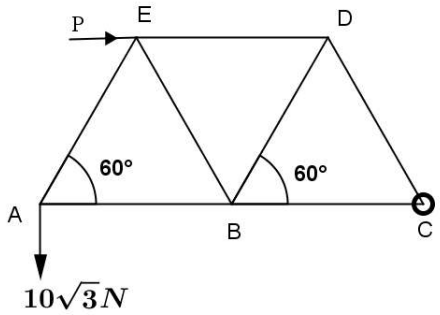
$R = \sqrt{40^2 + \left(10\sqrt{3}\right)^2}$ N

$R = 10\sqrt{19}$ N



Rod	Stress	Thrust	Tension
AB	$1000\sqrt{6}$ N	-	✓
BC	$1000\sqrt{3}$ N	-	✓
CD	2 000 N	✓	-
BD	$1000\sqrt{3}$ N	✓	-

Example 8



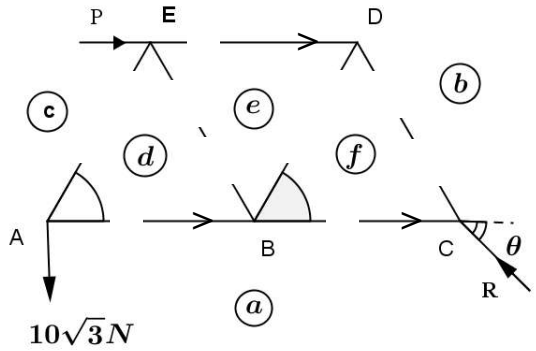
The given figure shows a framework consisting of seven light rods AB, BC, CD, DE, EA, EB and BD smoothly jointed at their extremities. The frame smoothly jointed at C and carries a load $10\sqrt{3}$ N at A. A horizontal force P at E keeps ED horizontal and the frame is in a vertical plane.

(i) Find the value of P at E

(ii) Find the magnitude and the direction of the reaction at C

(iii) Draw a stress diagram using Bow's notation and hence find the stresses in each rod distinguishing between tensions and thrusts.

(iv) From stres diagram varify reaction at C



For equilibrium

Take moment about C

$\text{n}\ P.\ell\cos 30^0 - 10\sqrt{3}.2\ell = 0$, where ℓ is length of a rod.

$\Rightarrow P = 40$N

Resolve the forces in the horizontal direction

$$\rightarrow \; P = R\cos\theta = 40N$$

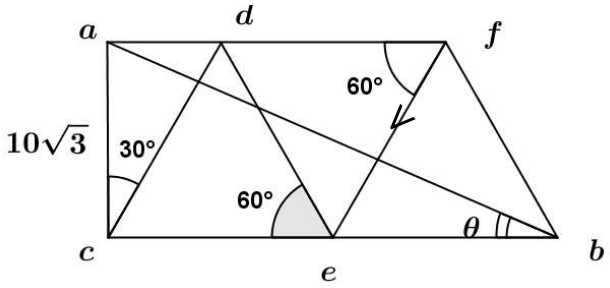
Resolve vertically

$$\uparrow R\sin\theta = 10\sqrt{3}N$$

$$R = \sqrt{40^2 + \left(10\sqrt{3}\right)^2}$$

$$R = 10\sqrt{19}N$$

$$\tan\theta = \frac{10\sqrt{3}}{40} \quad \Rightarrow \quad \theta = \tan^{-1}\left(\frac{\sqrt{3}}{4}\right)$$



Stress diagram

Start from joint A clockwise

Joint	order	Name of Polygon
A	$c \rightarrow a \rightarrow d \rightarrow c$	Δcad
E	$c \rightarrow d \rightarrow e \rightarrow b \rightarrow c$	$\square cdeb$
D	$b \rightarrow e \rightarrow f \rightarrow b$	Δbef

Join af, bf

AB = ad = $10\sqrt{3}\tan 30° = 10$ N

AE = cd = $10\sqrt{3}\sec 30° = 20$ N

cd = de = 20 N

AE = BE = 20 N

bf = de = ef = df = 20 N

CD = DE = BD = 20 N

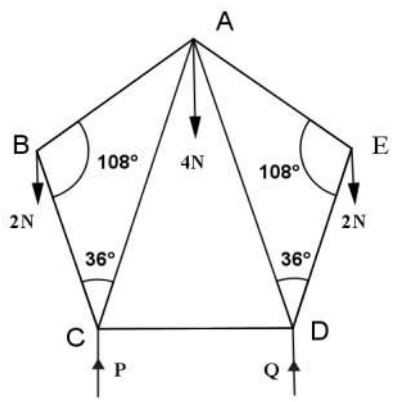
Reaction at c denoted by ab

$$ab^2 = \left(10\sqrt{3}\right)^2 + 40^2$$

$$ab = 10\sqrt{19}$$

Rod	Stress	
AB	10 N	Thrust
BC	30 N	Thrust
CD	20 N	Thrust
DE	20 N	Thrust
EA	20 N	Tension
EB	20 N	Thrust
DB	20 N	Tension

Example 9



A framework of seven freely jointed light rods is in the form of a regular pentagon ABCDE and the diagonals AC and BD. The framework is in the vertical plane with the lowest rod CD horizontal and is supported at C and D by two upward vertical forces of magnitude P and Q and weights 2 N, 4 N, 2 N are suspended at B, A and E respectively. Draw a stress diagram for this framework using Bow's notation. Hence determine the stresses in all seven rods, distinguishing between tensions and thrusts. Give the answers in terms of $\cos\frac{n\pi}{10}$ where n is a positive integer.

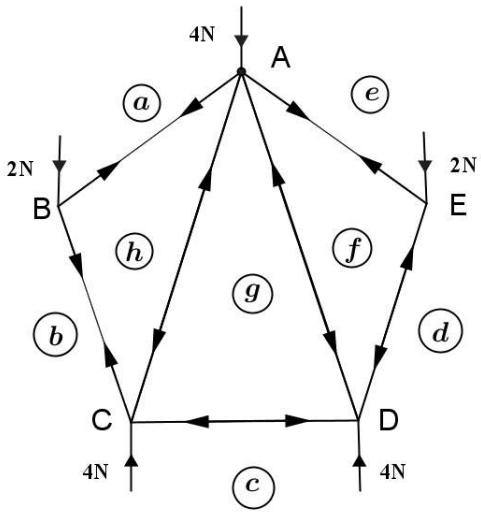
For the equilibrium of the system

Resolve the forces vertically

$$P + Q = 8 \text{ N}$$

$$\uparrow \quad P = a \qquad \text{(symmetry)}$$

$$P = Q = 4 \text{ N}$$



The system is symmetrical about the vertical line through A.

Start from the joint B and move in the clockwise direction.

$$n = 18° \ (\text{ say }) = \frac{\pi}{10}$$

de = ec = ca = ab = 2 N

let gc = x

Then pc = x tan 4n°

AB(bc) = Tension $= 100\sqrt{3}$ N

BC(ca) = Thrust $= 200\sqrt{3}$ N

CA(cd) = Tension $= 200\sqrt{3}$ N

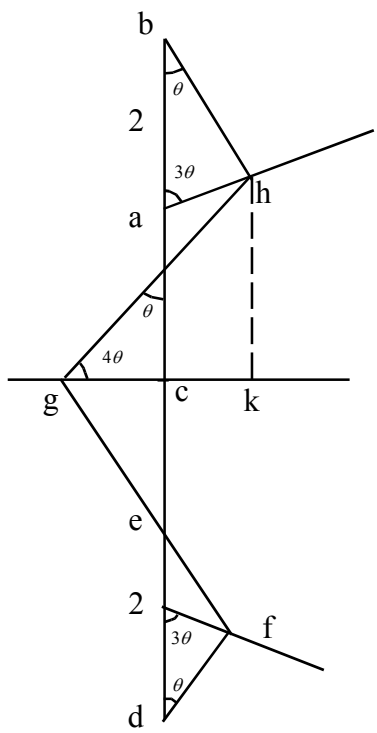
W(ad) = 200 N

First draw a vertical line and denote the vertical forces, in clockwise sense as

- ba, ae, ed, dc, ca

$$\theta = \frac{\pi}{10} = 18^0$$

Joint	Order	Polygon
B	b→a→h→b	Δ bah
E	e→d→f→e	Δ edf
A	h→a→e→f→g→h	⬠ haefg
C	b→h→a→c→h	□ bhgc



In Δabh, Using sine rule

$$\frac{ah}{\sin\theta} = \frac{2}{\sin 4\theta} = \frac{bh}{\sin 3\theta}$$

$$ah = 2\frac{\sin\theta}{\sin 4\theta},$$

$$bh = 2\frac{\sin 3\theta}{\sin 4\theta}$$

Δ 's abh and def are concruent

$$\therefore af = ah = \frac{2\sin\theta}{\sin 4\theta}$$

$$\therefore df = bh = \frac{2\sin 3\theta}{\sin 4\theta}$$

In Δ ghk

$$hk = gh\sin 4\theta = ac + ah\cos 3\theta$$

$$gh = \frac{2}{\sin 4\theta} + 2\frac{\sin\theta}{\sin 4\theta}.\cos 3\theta$$

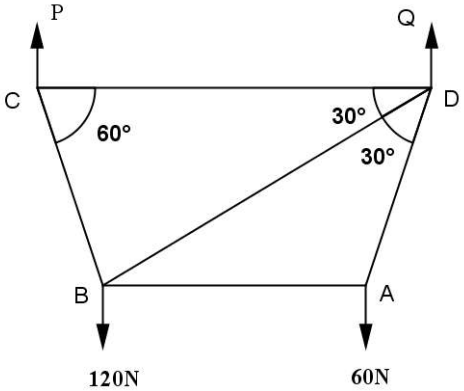
$$gc + ah\sin 3\theta = gh\cos 4\theta$$

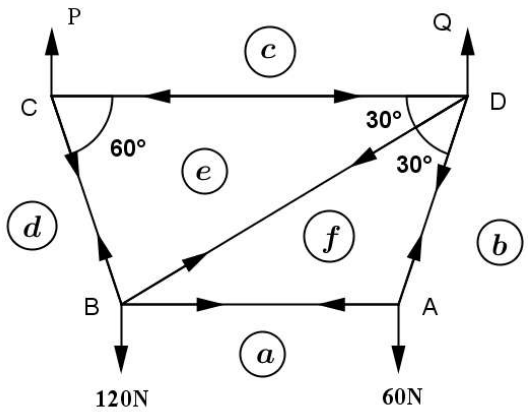
$$gc = gh\cos 4\theta - ah\sin 3\theta$$

Rod	stress	Tension	Thrust
AB	ah	✓	-
BC	bh	-	✓
AE	ah	✓	-
ED	bh	-	✓
AC	gh	-	✓
AD	gh	-	✓
DC	gc	✓	-

Example 10

The given frame consists of five light rods AB, AD, BC, BD and CD smoothly jointed at their extremities while the frame carries 120 N and 60 N at B and A respectively, the vertical forces P Newton and Q Newton applied at C and D to make AB and CD horizontal. Draw a stress diagram using Bow's notation and hence find the stresses in all five rods distinguishing between thrusts and tensions.





For equilibrium

Resolving vertically

$$\uparrow P + Q - 120 - 60 = 0$$

$$P + Q = 180N$$

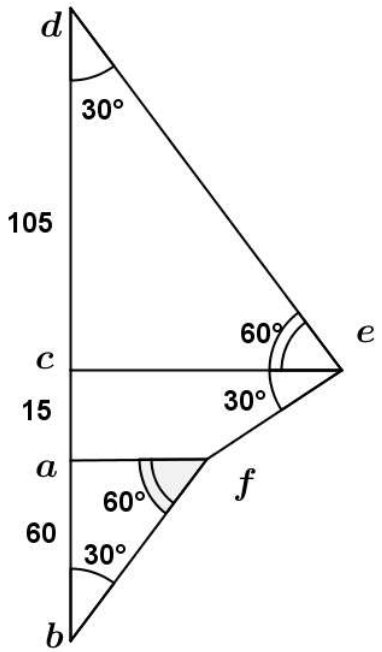
Taking moments about D

$$P.\,2\ell - 60.\ell\cos 60 - 120.\ell + \ell\cos 60 = 0$$

$$2P = 30 + 180$$

$$\Rightarrow P = 105N$$

$$\Rightarrow Q = 180 - 105 = 75N$$



Stress diagram

Start from joint C and move anticlockwise

C → B → A

Step I : Cm Step II : Bm Step III : Am

$AB = af = 60 \tan 30° = 20\sqrt{3}$ N

$BC = ed = 105 \sec 30° = 70\sqrt{3}$ N

$CD = ec = 105 \tan 30° = 35\sqrt{3}$ N

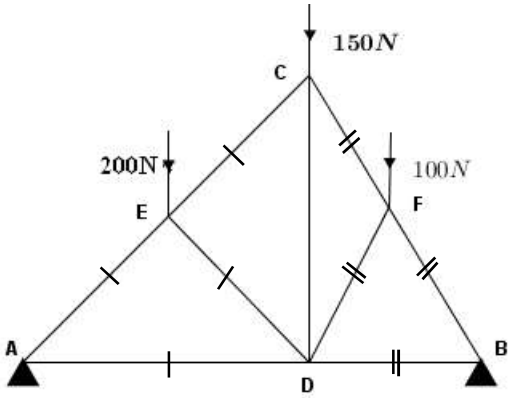
$AD = bf = 60 \sec 30° = 40\sqrt{3}$ N

$BD = ef = 15 \sec 30° = 10\sqrt{3}$ N

Rod	Stress	
AB	$20\sqrt{3}$ N	Tension
BC	$70\sqrt{3}$ N	Tension
CD	$35\sqrt{3}$ N	Thrust
AD	$40\sqrt{3}$ N	Tension
BD	$10\sqrt{3}$ N	Tension

6.4 Exercises

1.

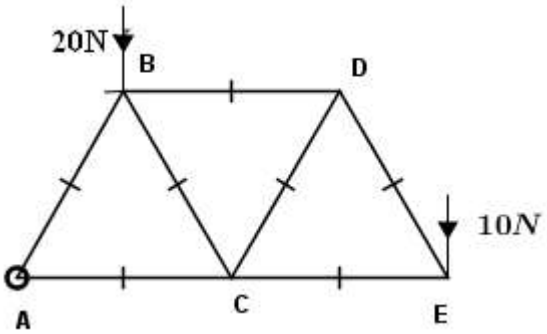


The figure represents the framework of a roof whose weight may be regarded as distributed in the manner shown above.

i. Find the reaction at A and B.

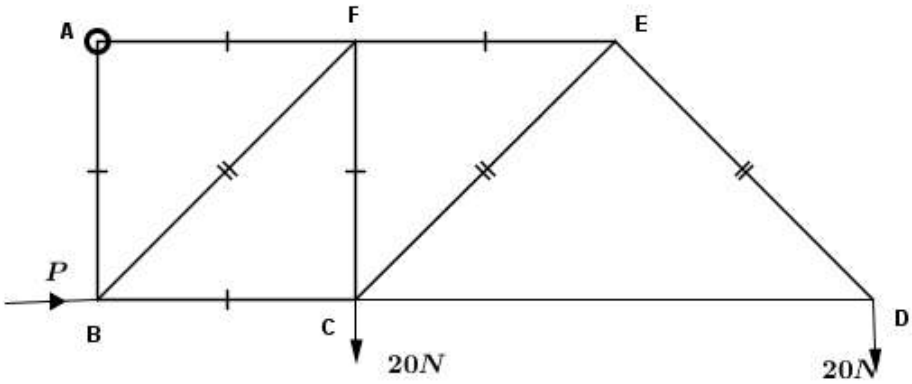
ii. Draw the stress diagram by using Bow's notation and find the stress in each rod, distinguishing between tensions and thrusts.

2.



The above figure shows a framework made by seven light rods. The frame is hinged at A to a fixed point and kept in position by a horizontal force P at B. Draw a stress diagram using Bow's notation and find the stress in each rod distinguishing between tensions and thrusts.

3.

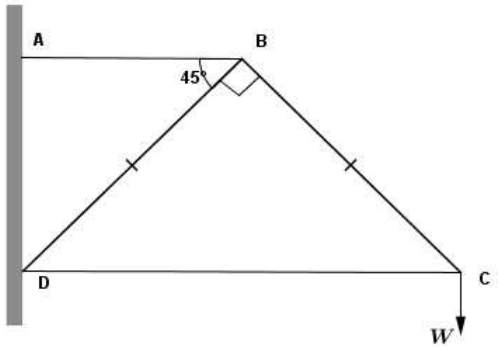


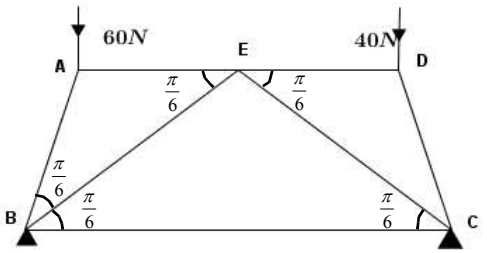
The above framework consists of nine smoothly jointed light rods, smoothly hinged to a fixed point at A, kept in equilibrium by a horizontal force P at B and loaded with 20 *W* each at C and D.

i. Find P and the reaction at A.

ii. Draw the stress diagram using Bow's notation and find the stress in each rod, distinguishing between tensions and compressions.

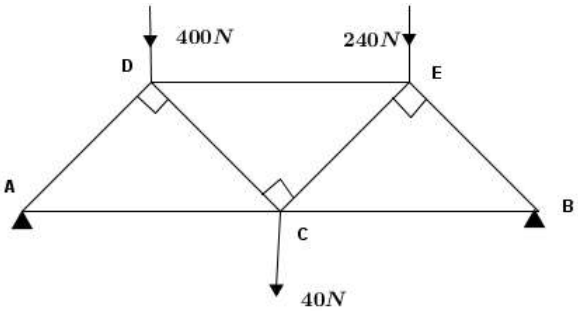
4. The framework consists of four light rods AB, BC, CD and DB freely jointed at B, C, D and attached to a vertical wall at A, and D loaded with WN at C. Draw a stress diagram using Bow's notation and find the stress in each rod, distinguishing between tensions and thrusts.

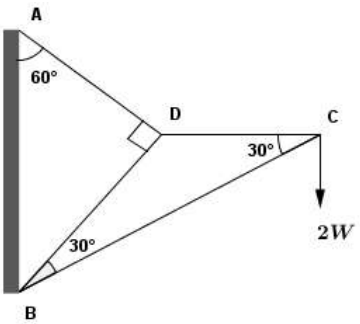


5. 

ABCDEF is a framework which has freely jointed rods AB, BC, CD, DE, AE, BE and CE such that $E\hat{B}C=E\hat{C}B=A\hat{B}E=D\hat{C}E=A\hat{E}B=D\hat{E}C=\frac{\pi}{6}$. The framework is supported at B and C such that BC is horizontal, and loaded 60 N, 40 N at A and D respectively. Draw a stress diagram using Bow's notation and find the stress in each rod, distinguishing thrusts and tensions.

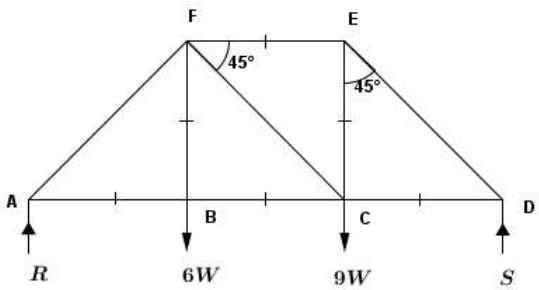
6. The framework in the figure is formed by using light bars according to the diagram. All triangles are right angular and isosceles. The system is on support at A and B such that ACB horizontal. The framework carries loads of 40 N, 400 N and 240 N at C, D and E respectively. Draw stress diagram using Bow's notation and find the stresses in the bars distinguishing tensions and thrusts.



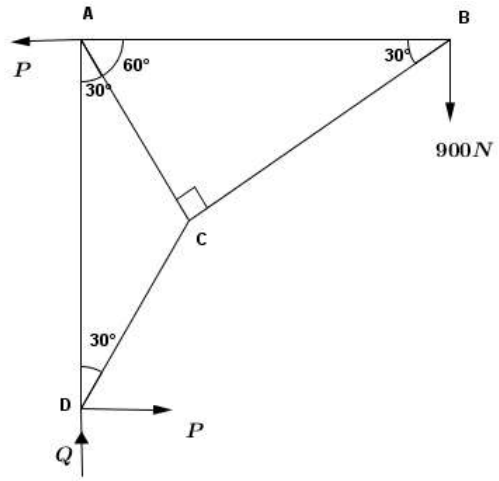
7. 

The figure shows a framework consisting freely jointed four light bars AD, BD, BC and CD. It is hinged freely to a vertical wall at A and B. C carries a load of $2W$. By using stress diagram find the reactions at A and B. Hence find the stresses in the rods distinguishing tensions and thrusts.

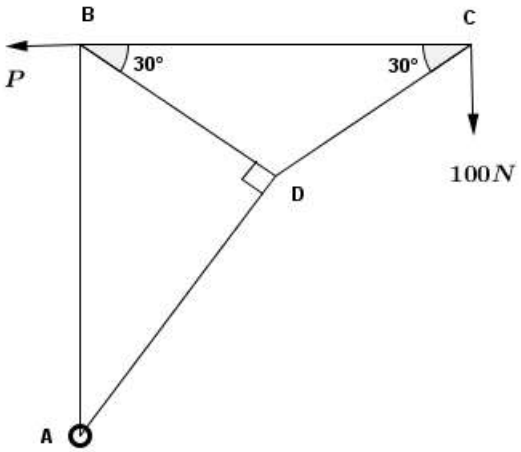
8. The figure shows a framework consisting nine light rods freely jointed at A, B, C, D, E and F. The frame carries loads of $6W$ and $9W$ at B and C respectively. It is supported by vertical forces R and S at A and D respectively. Draw a stress diagram and find the stresses in the rods distinguishing between tensions and thrusts.



9. A freely jointed framework consisting of five light rods is shown in the figure. Joint B carries a load of 900N. The framework is in equilibrium such as AD is vertical by means of forces P and (P, Q) acting on A and D respectively (P is horizontal and Q is vertical). Find the magnitudes of forces P and Q. Draw the stress diagram using Bow's notation and find the stress in each rod, distinguishing tensions and thrusts.



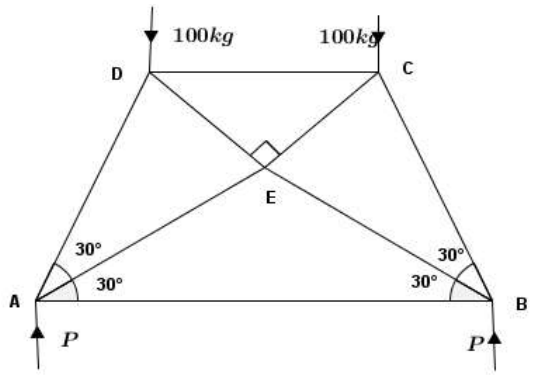
10.



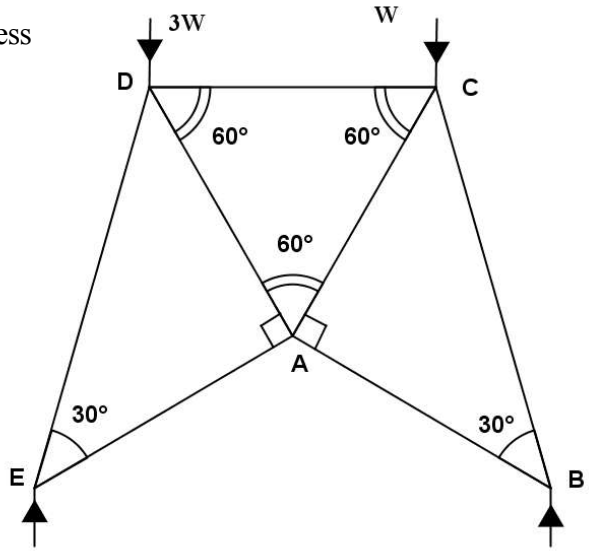
Five light rods are freely jointed to form the framework shown in the above figure. The framework is in equilibrium in a vertical plane with joint A freely hinged to a fixed point. AB is vertical, BC is horizontal, $A\hat{D}B = 90°$ and $D\hat{B}C = D\hat{C}B = 30°$. A load of 100 N hangs at C and a horizontal force P acts at B in the direction of CB.

Find P and obtain the horizontal and vertical components of the reaction on the hinge at A. Draw a stress diagram for the framework using Bow's notation. Hence determine the stresses in all five rods distinguishing tensions and thrusts.

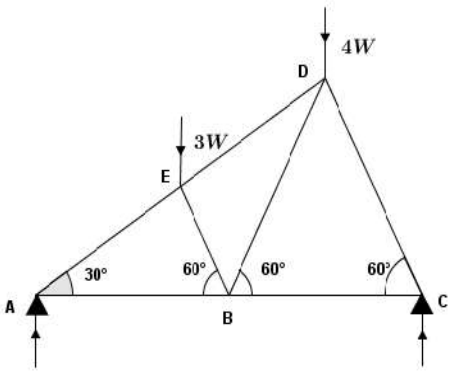
11. The given framework consists freely jointed eight light rods at A, B, C, D and E. The joints A and B are on vertical supports P at each joint. The framework carries equal loads of 100 kg at points C and D. AB is horizontal and AE=BE=AD=BC. Find the value of P. Assuming the thrust in C as X kg draw a stress diagram for the framework. If the tension on AB is Y kg, using the geometry of the stress diagram, prove that $y = 100 - (\sqrt{3} - 1)X$. Explain why the real values of x and y cannot be calculated simultaneously. Find the stress in every rod if X = Y.



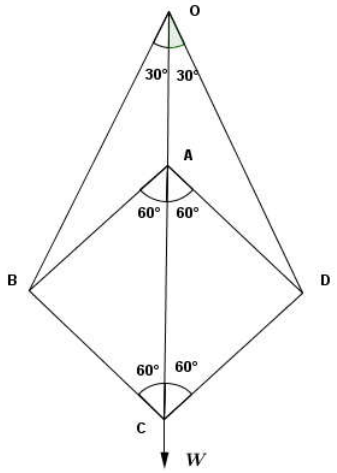
12. The given figure represents a framework which is formed by seven light rods. Ends A, B, C, D, E are freely jointed. This framework carries loads W and $2W$ at joints C and D, and is supported at B and E such that BE is horizontal. Draw a stress diagram using Bow's notation and find the stress in every rod distinguishing tensions and thrusts.



13. The framework consisting seven freely jointed light rods is placed on two supports at A and C such as the framework carries loads $4W$ and W at D and E respectively. Find the reactions at A and C. Find the stresses in each and every rod using a stress diagram, distinguishing between them compressions and tensions.

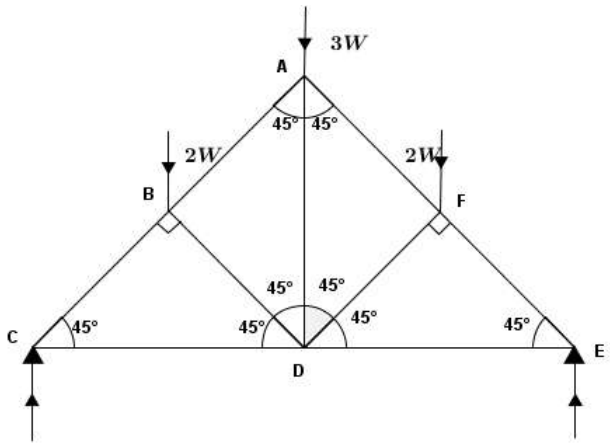


14.

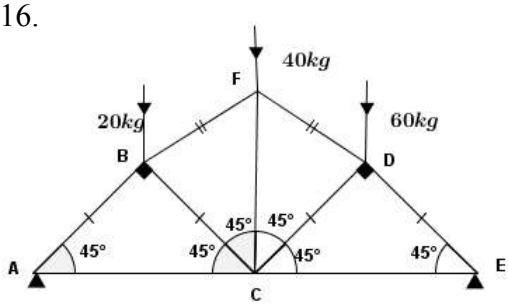


The given framework is a rhombus formed by freely jointed five light rods. It is hung from A by means of equal strings OB, OD, and OA is a vertical rod freely jointed at A. The diagonal AC of the rhombus is vertical and $A\hat{B}C = B\hat{O}D = 60°$. When C carries a load W, find the stresses in each rod and also the tensions in the strings by using a stress diagram. Name the rods which are under tension.

15. The figure shows a framework formed by freely jointed light bars. DA is vertical. The framework is supported at C and E. It carries loads $3W$, $3W$ and W at joints A, B and F, respectively. Find the reactions at C and E. Draw a stress diagram using Bow's notation and find the stresses in each rod. Distinguish thrusts and tensions.



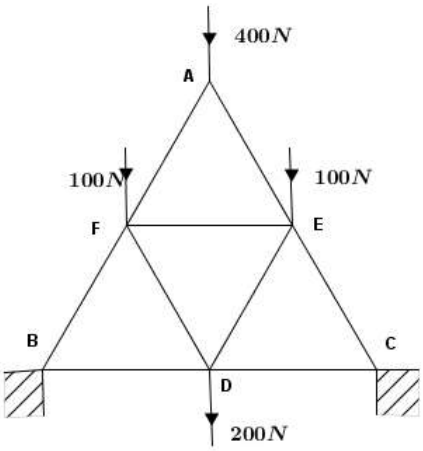
16.



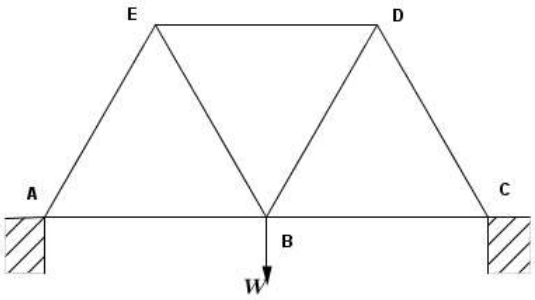
The given figure represents a framework of light bars loaded at joints B, F, D as indicated. The bars AC and CE are horizontal and each equal to 10 m and CF = 8 m. Also the lengths AB = BC = CD = DE and BF = FD. The frame rests on two smooth pegs at A and E. Calculate the reactions at A and E assuming that they are vertical. Draw a stress and find the stresses in the rods distinguishing between tensions and thrusts.

17. The given framework formed by nine equal light bars, carries loads as shown in the figure. The framework is at rest on B and C on supports such that the system is in a vertical plane.

Find the reaction at B and C. Draw a stress diagram by using Bow's notation. Hence calculate the stresses in each rod distinguishing tensions and thrusts.



18. The framework of a bridge ABCDE which is formed by seven light equal rods is shown in the figure. The joints A and C are on supports which are in same horizontal level and the framework is in a vertical plane and B carries a load W. Draw a stress diagram using Bow's notation. Hence find the stress in each rod distinguishing thrusts and tensions.

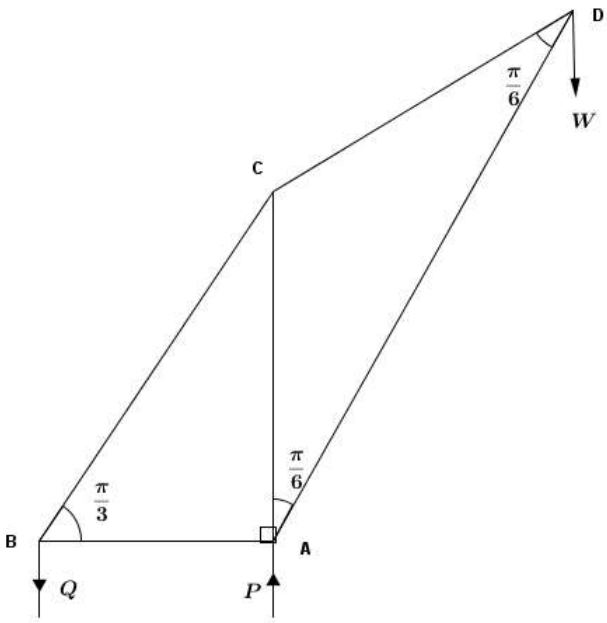


19. The framework consisting light rods AB, BC, CA, CD and DA are freely jointed at their extremities is placed in a vertical plane with AB horizontal and AC vertical, AB = a, $B\hat{C}D = B\hat{A}D = \frac{2\pi}{3}$ and $A\hat{B}C = \frac{\pi}{3}$. The framework supports a vertical load W at D and the equilibrium is maintained by two vertical forces P, Q at A and B respectively.

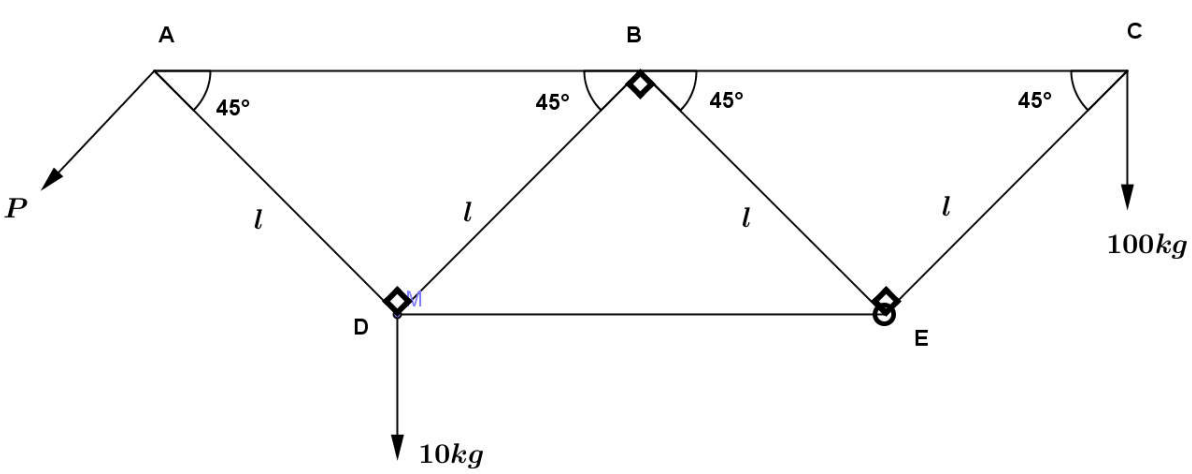
(i) Find P and Q in terms of W

(ii) Draw a stress diagram for this framework using Bow's notation.

Hence determine the stresses in the five bars distinguishing thrusts and tensions.



20.



The above framework is made by seven light rods AB, BC, AD, BD, BE, CE and DE where AD = BD = BE = CE = ℓ. The frame is hinged at E, kept in equilibrium by a force P applied at A, with loads 100 kg at C and 10 kg at D.

(i) Find the vertical and horizontal components of the reaction at E.

(ii) Find the value of P.

(iii) Draw a stress diagram using Bow's notation and hence find stresses in each rod distinguishing tensions and thrusts.